Naphat Jakkraphatcharakul

Bangkok, Thailand | Available for internship/co-op: January - May 2027 (4 months)
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EDUCATION

Thammasat University **2023 - Present** **Expected 2027**
B.Sc. in Data Science and Innovation (Cybersecurity module), College of Interdisciplinary Studies | GPA 3.33/4.00

WORK EXPERIENCE

Business Administration Intern **UMEDA Co., Ltd. (Bangkok, TH)** **June 2026 - August 2026**

- Built Excel workbooks that break down pharmaceutical sales by manufacturer and product across government, private and OTC channels, grouped by ATC3 therapeutic class, and ranked both within each channel.
- Compared hospital and reference prices, then projected future prices with Excel trend and forecast tools.
- Turned the rankings and price comparisons into Power BI dashboards and PowerPoint decks for the Marketing team's meetings with executives.

PROJECTS

Portfolio website: https://serenaurity.github.io/ (for additional information and projects)

Thrive - Rehabilitation Data Analysis **January 2026 - Present**
Academic project for the Ministry of Justice's Kamlangjai program | Python, SQL (DuckDB), Parquet, scikit-learn

- Linked 12 corrections datamart tables with probation data in DuckDB into person-level Parquet tables covering 2M+ people, with a data contract that tags each feature as usable, proxy or protected.
- Compared occupational-grant recipients with non-recipients using direct standardization by sex, age band and prisoner class, applied Benjamini-Hochberg correction across all tested factors, and presented the results in an aggregate-only dashboard.
- Benchmarked multinomial logistic, ordinal and gradient-boosting models against a majority baseline for prisoner class (levels 1-6); found the top features were inputs to the official classification rule, so reported the models as a rule-consistency check, backed by a fairness audit and model card.

CVE-to-ATT&CK Technique Ranking **September 2026 - Present**
Academic Research Project | Python, LightGBM (LambdaRank), BM25, MiniLM (sentence-transformers), pandas

- Built a CVE-CWE-CAPEC-CPE knowledge graph that keeps the source and date of every link, and trained a LambdaRank model on 44 text and graph features to rank MITRE ATT&CK techniques for each CVE role.
- Under a pre-registered protocol, reached nDCG@5 0.558 on a sealed test set (158 queries) versus 0.218 for MiniLM embeddings and 0.145 for BM25; 5×5-fold CV ablations traced most of the gain over text-only to platform (CPE) features, as a randomized graph scored the same as the real one

Student Dropout Risk Research **August 2025 - December 2025**
Academic Research Project

- Cleaned course-level records for 20,919 undergraduates (2014-2024), rolled them up to credit-weighted term GPA, and engineered GPA-trend, variability and course-failure features.
- Compared logistic regression, random forest and XGBoost at three checkpoints with a student-level 80/20 split. XGBoost led on ROC-AUC and F1 at every stage: ROC-AUC 0.866 after Term1, rising to 0.992 after Term3.

PUBLICATION

A Machine Learning-Based Student Dropout Risk Prediction and Early Warning System. Asian Health, Science and Technology Reports, 34(4), 2026 (accepted). Second author; presented at AJCC 2026, Sapporo, Japan.

SKILLS

Programming: Python (pandas, scikit-learn, XGBoost), SQL (DuckDB, MySQL), NoSQL (MongoDB), TypeScript
Statistics & Machine Learning: EDA, hypothesis testing, direct standardization, multiple-testing correction, regression, classification, learning to rank, NLP (text embeddings), model evaluation, fairness audit
Data engineering: data modeling, ETL/ELT, data warehousing, data quality checks, data governance, data contracts
BI & analytics: Power BI, Tableau, Excel (pivot tables, forecasting), KPI and dashboard design, pricing analysis
Tools & security: Git, Docker (web project), Linux, digital forensics (Volatility), log analysis, network security
Languages: Thai (Native), English (Upper Intermediate)
TU-GET 550/1000 (Reading 330/500, Vocabulary 130/250, Structure 90/250)